

Worksheet on Chapter 8 Ch 8.1, 8.2, 8.3

8.1 Point and confidence interval estimates for a population mean

8.2 Construct a 90% confidence interval estimate for the population mean given the following values:

$$\bar{x} = 70, \sigma = 15, n = 65$$

8.3 Construct a 95% confidence interval estimate for the population mean given the following values:

$$\bar{x} = 300, \sigma = 55, n = 250$$

8.6 Determine the margin of error for a confidence interval estimate for the population mean of a normal distribution given the following information:

b. confidence level = 0.99, $n = 25$, $\sigma = 3.47$

c. confidence level = 0.98, standard error = 2.356

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8.11 A random sample was selected from a population having a normal distribution. Calculate a 90% confidence interval estimate for μ for each of the following situations:

b. $\sum x = 40.5$, $n = 9$, $\sigma = 2.9$

8.15 Car buyer often add accessories to their cars. A sample of 179 recent Mini Cooper purchasers yielded a sample mean of \$5000 above the \$20200 base sticker price. Suppose the cost of accessories purchased for all Mini Coopers has a standard deviation of \$ 1,500.

- Calculate a 95% confidence interval for the average cost of accessories on Mini Coopers.
- Determine the margin of error in estimating the average cost of accessories on Mini Coopers.